

Torrefaction Heating Jacket

My role is to simply provide a certain amount of energy to the surface of the wall. Everything is simulated as if the heating jacket is concurrent pipes.

```
> restart;

> print("====Key Equations====");
eq1 := Q__req = m*Cp*(T__hot1 - T__hot2); #Energy Balance from
page 478 fundametals of heat and mass transfer
eq2 := Q__req = h__j*A*DeltaT__lm;          #Page 483 fundametals
of heat and mass transfer
DeltaT__lm := ((T__hot1 - T__cold)-(T__hot2 - T__cold))/(ln(
(T__hot1-T__cold)/(T__hot2-T__cold))); #Page 482 fundametals of
heat and mass transfer
R__gas := 8.314:
```

$$\begin{aligned} & \text{"====Key Equations===="} \\ eq1 &:= Q_{req} = m C_p (T_{hot1} - T_{hot2}) \\ eq2 &:= Q_{req} = h_j A \Delta T_{lm} \\ \Delta T_{lm} &:= \frac{T_{hot1} - T_{hot2}}{\ln \left(\frac{T_{hot1} - T_{cold}}{T_{hot2} - T_{cold}} \right)} \end{aligned} \quad (1.1)$$

```
> #1 is in
#2 is out
#Hot is oil/salt/water
#Cold is the wall
```

```
print("====Specifications====");
T__hot1 := 282.0064732 + 273.15; #Kelvin (Temperature out of air
Heater) (Assumes no loss of energy in pipes)
m := 0.25; #kg/s (initial calculation from Air heater)

T__cold := 280 + 273.15; #Kelvin #depends on energy for pyrolizer

D__pyrolizer := 0.50; #m #Diameter of Torrefaction
```

```

D__jacket := 0.552; #m #Diameter of heating jacket
L := 4.2; #m #Length of pyrolizer

Q__req := 11290; #W #Required energy for Torrefaction

```

```

print ("====Properties of Helisol 5A, tube side====");
rho := 689.5; #kg/m3
Cp := 2150; #J/kg/K
F_mu__salt := 0.41e-3; #Pa/s
k__fluid := 0.076 #W/m/K

```

"====Specifications===="

$T_{hot1} := 555.1564732$

$m := 0.25$

$T_{cold} := 553.15$

$D_{pyrolizer} := 0.50$

$D_{jacket} := 0.552$

$L := 4.2$

$Q_{req} := 11290$

"====Properties of Helisol 5A, tube side===="

$\rho := 689.5$

$C_p := 2150$

$F_{\mu_{salt}} := 0.00041$

$k_{fluid} := 0.076$

(1.2)

> #helical cannels (If pressure is too high, consider splitting channels into two, but will half velocity and Re.)

w := 0.062; #m #Channel width

t := 0.330; #m #Wall thickness

P := w + t; #m #pitch

N__turn := L/P; #number of turns

L__turn := ((Pi*D__coil)**2 + P**2)**(1/2); #m #Length per turn

L__coil := N__turn * L__turn; #m

```

GAP := (D__jacket - D__pyrolizer)/2;
A__c := GAP*w;
D__h := 2*GAP*w/(GAP +w);
D__coil := (D__jacket + D__pyrolizer)/2;

```

$w := 0.062$

$t := 0.330$

$P := 0.392$

$N_{turn} := 10.71428571$

$L_{turn} := \sqrt{\pi^2 D_{coil}^2 + 0.153664}$

$L_{coil} := 10.71428571 \sqrt{\pi^2 D_{coil}^2 + 0.153664}$

$GAP := 0.02600000000$

$A_c := 0.001612000000$

$D_h := 0.03663636364$

$D_{coil} := 0.52600000000$

(1.3)

```

> T__hot2 := solve(eq1,T__hot2); #kelvin
T__hot_avg := (T__hot2 + T__hot1)/2;
mu := F_mu__salt(T__hot_avg);
v := m/(rho*A__c); #m/s
A := Pi * D__pyrolizer*L;

```

$T_{hot2} := 534.1518220$

$T_{hot_avg} := 544.6541476$

$\mu := 0.00041$

$v := 0.2249265390$

$A := 6.597344573$

(1.4)

```

> Re__1 := rho*v*D__h/mu;

```

$\Re_l := 13858.09312$

(1.5)

```

> print("====Summary====");
Re__1 := Re__1;
v := v; #m/s
T__hot1 := T__hot1; #Kelvin
T__hot2 := T__hot2; #Kelvin
print("=====");

```

"====Summary===="

$\Re_l := 13858.09312$

```

v := 0.2249265390
T_hot1 := 555.1564732
T_hot2 := 534.1518220
"=====

```

(1.6)

```
> #Size CHECK
```

```

Pr := (Cp *mu)/k__fluid;
#cooling, Page 491, fundametals of heat and mass transfer
#Updated to use Seban and McLaughlin correlation
Nu := 0.023*Re__1**0.85*Pr**0.3*(D__h/D__coil)**0.1;
h__j := Nu*k__fluid/D__h; #heat transfer coefficient (Book uses
U)

```

```
Pr := 11.59868421
```

```
N := 121.8403307
```

```
h_j := 252.7506612
```

(1.7)

```
> dT1 := T__hot1 - T__cold;
dT2 := T__hot2 - T__cold;
DeltaT_lm := (dT1 - dT2) / ln(dT1 / dT2);
```

```

# Required Area vs Available Area
A__req := Q__req / (h__j * DeltaT__lm);
A__av := 3.14 * D__pyrolizer * L;

```

```

# Check if design is valid
if A__av > A__req then
    print("Design Successful");
else
    print("Design Failed");
end if;

```

```
dT1 := 2.0064732
```

```
dT2 := -18.9981780
```

```
DeltaT_lm := -3.164100544 - 4.421918101 I
```

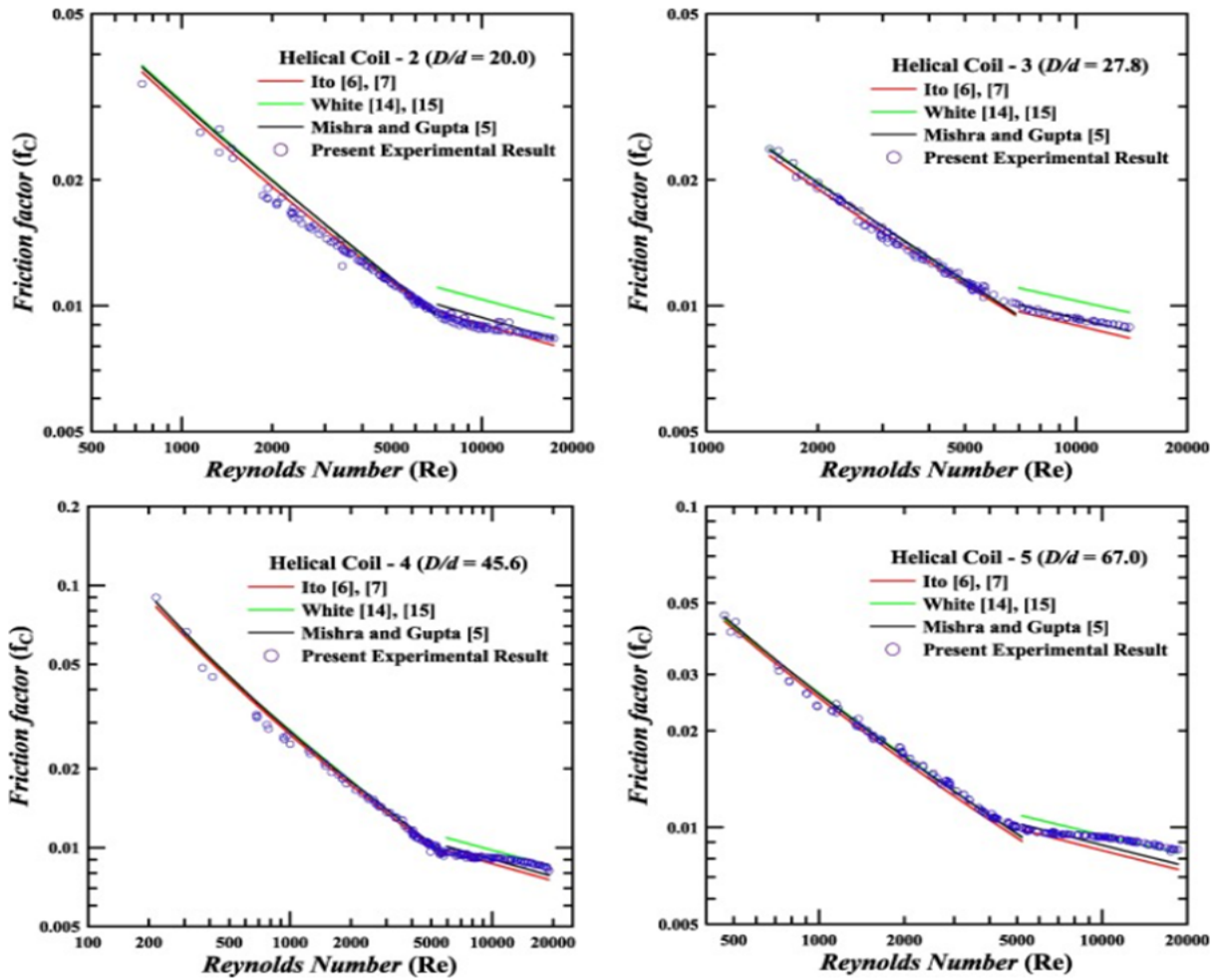
```
A__req := -4.780525299 + 6.680916443 I
```

```
A__av := 6.59400
```

```
"Design Failed"
```

(1.8)

```
> #Pressure calculations (https://www.sciencedirect.com/science/article/pii/S0017931015005633) [I belive paper uses fanning friction factor]
GraphDeterminer := D__coil / D__h;
ReferenceReynoldsNumber := Re__1;
```



$$GraphDeterminer := 14.35732010$$

$$ReferenceReynoldsNumber := 13858.09312$$

(1.9)

> $f_c := 0.009$; #ALWAYS NEED TO MEASURE PROPERLY

$$f_c := 0.009$$

(1.10)

> #Assumes no added pressure from coiling up and down sides, gravity pressure is negligible.

$$DP := 4 * f_c * (L_{coil} / D_h) * ((\rho * v^2) / 2); \text{ #Pa}$$

$$DP_{kPa} := DP / 1000; \text{ #kPa}$$

$$DP := 311.8622210$$

$$DP_{kPa} := 0.3118622210$$

(1.11)